

Treat the following questions:

1. Describe:
 - a. The client-server architecture 2-tier and 3-tier. Compare between the two architectures. Draw a schema for each showing its functionality (3 pts)
 - b. Software architecture centered on events. (event-driven architecture) (1 pt)
2. Talk about the different specifications techniques. (4 pts)
3. The prototype model: (4 pts)
 - a. Definition
 - b. Necessity
 - c. The different types

4. Petri Nets

We want to model the management of cabins and baskets for entrance to a swimming pool. When entering, a person who finds a free cabin changes his clothes in the cabin, then requests a basket to fill with clothes in order to leave the cabin. After swimming, the person enters a cabin with his basket, empties it, liberates the basket, and then gets dressed to liberate the cabin.

Model this organization using Petri Nets with a place (token) for the stock of cabins (3 for example), and another place (token) that models the stock of baskets (5 for example). What is the maximum of simultaneous swimmers? Show that this organization risks having a blocking issue. (4 pts)

5. White Box Test

Having the following string comparison program expressed in pseudo-code:

1. *equal* : a Boolean
2. *string1*, *string2* : 2 character strings
3. *Read(string1, string2)*
4. *If (string1.length = string2.length) Then* // *string1.length* gives the length of *string1*
5. *i = 1*
6. *While i <= string1.length AND string1.character[i] = string2.character[i]* {
// *string1.character[i]* gives the *i*th character of *string1*
7. *i = i + 1* }
8. *If i = string1.length + 1 then* // the first character has an index 1.
9. *Print(« strings are equals »)*
10. *Else Print(« strings are not equal »)*

Give a set test covering all possible paths; assuming that the string has a size ≤ 2 (0, 1, or 2). To manage, draw the Control-Flow Graph, number the nodes of the graph then list all the paths with the sequence of numbers (4 pts)